Python Programming - 2301CS404

Lab - 4

Roll No.: 496

Name: Dhruv Limbasiya

24031701027

01) WAP to print 1 to 10.

In [2]:
```
for i in range(1,11):
    print(i)
```

```
1
2
3
4
5
6
7
8
9
10
```

02) WAP to print 1 to n.

In [3]:
```
lastNum = int(input("Enter Last Number :"))
for i in range(1,lastNum+1):
    print(i)
```

```
1
2
3
4
5
```

03) WAP to print odd numbers between 1 to n.

In [4]:
```
lastNum = int(input("Enter Last Number :"))
for i in range(1,lastNum+1):
    if i%2!=0:
        print(i)
```

```
1
3
5
```

04) WAP to print numbers between two given numbers which is divisible by 2 but not divisible by 3.

In [6]:
```
firstNum = int(input("Enter First Number :"))
lastNum = int(input("Enter Last Number :"))
for i in range(firstNum,lastNum+1):
    if (i%2==0) and (i%3!=0):
        print(i)
```

```
2
4
8
10
14
```

05) WAP to print sum of 1 to n numbers.

In [7]:
```
num = int(input("Enter Number :"))
sum=0
for i in range(1 , num+1):
    sum+=i
print(f"Sum to {num} digits = {sum}")
```

```
Sum to 5 digits = 15
```

06) WAP to print sum of series 1 + 4 + 9 + 16 + 25 + 36 + ...n.

In [12]:
```
lastNum = int(input("Enter Last Number :"))
sum = 0
for i in range(1,lastNum+1):
    sum+=(i*i)
    if i == lastNum:
        print(i*i ,end="=")
    else:
        print(i*i, end="+")
print(sum)
```

```
1+4+9+16+25+36=91
```

07) WAP to print sum of series 1 – 2 + 3 – 4 + 5 – 6 + 7 ... n.

In [20]:
```
lastNum = int(input("Enter Last Number :"))
sum = 0
for i in range(1,lastNum+1):
    if i == lastNum:
        if i%2!=0:
            sum+=i
        else:
            sum-=i
        print(i ,end="=")
    elif i%2!=0:
        sum+=i
        print(i, end="-")
    else:
        sum-=i
        print(i,end="+")
print(sum)
```

```
1-2=-1
```

08) WAP to print Multiplication Table of the given number.

In [22]:
```
num = int(input("Enter Number For Multiplication Table :"))
for i in range(1,11):
    print(f"{num} * {i} = {num*i}")
```

```
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
5 * 7 = 35
5 * 8 = 40
5 * 9 = 45
5 * 10 = 50
```

09) WAP to find Factorial of the given number.

In [26]:
```
num = int(input("Enter Number For Factorial :"))
ans=1
for i in range(1,num+1):
    ans*=i
print(f"{num}! = {ans}")
```

```
5! = 120
```

10) WAP to print GCD of given two numbers.

In [32]:
```
num1 = int(input("Enter First Number :"))
num2 = int(input("Enter Second Number :"))
greatest = num1 if num1>=num2 else num2
for i in range(greatest ,0 ,-1):
    if num1%i==0 and num2%i==0:
        print(f"GCD of {num1} and {num2} is {i}")
        break
```

```
GCD of 12 and 18 is 6
```

11) WAP to find Factors of the given number.

In [27]:
```
num = int(input("Enter Number For Factor :"))
for i in range(1,num+1):
    if num%i==0:
        print(i)
```

```
1
2
3
6
9
18
```

12) WAP to find whether the given number is Prime or not.

In [39]:
```
num = int(input("Enter Number For Prime or not :"))
for i in range(2,num):
```

```
    if num%i==0:
        print(f"{num} is not prime.")
        break
else :
    print(f"{num} is prime.")
```

```
23 is prime.
```

13) WAP to print sum of digits of given number.

In [40]:
```
num = int(input("Enter Number For Sum of Digits :"))
temp = num
ans=0
while(temp>0):
    ans+=(temp%10)
    temp//=10
print(f"Sum of digit {num} = {ans}")
```

```
Sum of digit 123 = 6
```

14) WAP to check whether the given number is Palindrome or not.

In [42]:
```
num = int(input("Enter Number For Sum of Digits :"))
temp = num
ans= rev =0
while(temp>0):
    r=(temp%10)
    rev = rev*10 + r
    temp//=10

if num == rev:
    print(f"{num} is palindrome")
else:
    print(f"{num} is not palindrome")
```

```
121 is palindrome
```

15) WAP to check whether the given number is an Armstrong Number or not.

In [44]:
```
num = int(input("Enter Number For Sum of Digits :"))
temp = num
digits=ans=0
while(temp>0):
    digits+=1
    temp//=10

temp = num
while(temp>0):
    r = temp%10
    ans+=(r**digits)
    temp//=10

if num == ans:
    print(f"{num} is Armstrong")
else:
    print(f"{num} is not Armstrong")
```

```
153 is Armstrong
```

16) WAP to print all the perfect numbers between 1 to n.

In [5]:
```
num = int(input("Enter Last Number For Perfect Number list :"))
for i in range(1,num+1):
    sum =0
    for j in range(1,i):
        if i%j==0:
            sum+=j
    if sum == i :
        print(f"{i} is Perfect Number.")
```

```
6 is Perfect Number.
28 is Perfect Number.
```

In [ ]: